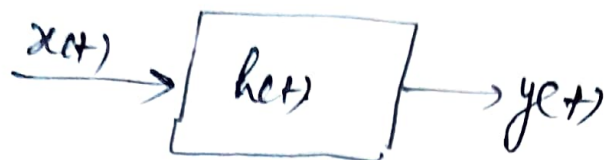


Transfer function of continuous time LTI system

①



$$y(t) = h(t) * x(t)$$

Taking Laplace on both side

$$Y(s) = H(s) X(s)$$

$$H(s) = \frac{Y(s)}{X(s)} \rightarrow \text{Called transfer function of the system}$$

→ Note:- Causality of continuous time LTI system

①

→ If the ROC of $H(s)$ is the ^{of right} rightmost pole, then system $H(s)$ will be causal, otherwise Noncausal

② Stability of continuous time LTI system

If the ROC include the imaginary axis ($j\omega$), the system is stable otherwise unstable.

Qe: A system is described by following differential eqⁿ

$$\frac{d^2 y}{dt^2} - \frac{dy}{dt} - 6y(t) = \frac{dx}{dt} - 2x(t)$$

Determine

- (a) Transfer function of the system
- (b) Poles-zero plot of the system
- (c) Determine the impulse response ~~h(t)~~ $h(t)$ and also specify the ROC if
- the system is causal
 - if the system is stable
 - if the system is non-causal & unstable.

Solⁿ:- (a) $\frac{d^2 y}{dt^2} - \frac{dy}{dt} - 6y(t) = \frac{dx}{dt} - 2x(t)$

Taking Laplace on both side

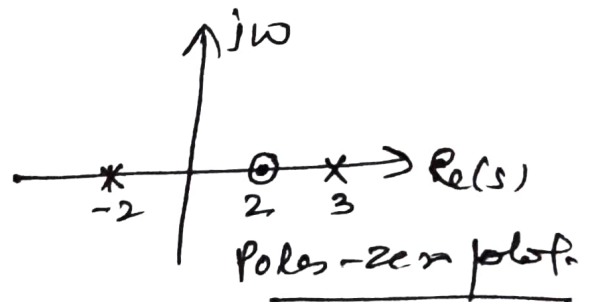
$$s^2 y(s) - s y(s) - 6y(s) = s X(s) - 2X(s)$$

$$H(s) = \frac{Y(s)}{X(s)} = \frac{(s-2)}{(s^2 - s - 6)} = \frac{(s-2)}{(s+2)(s-3)}$$

(b) Poles-zero plot

Poles:- $s = -2, 3$

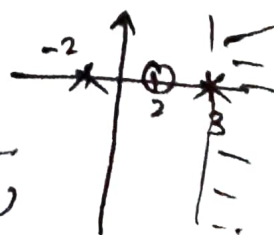
zero:- $s = 2$



(c) (i) The system is causal,
ROC should be right of right most pole of $H(s)$
i.e. $\text{Re}(s) > 3$

~~h(t)~~ $\Rightarrow H(s) = \frac{(s-2)}{(s+2)(s-3)} = \frac{4/5}{(s+2)} + \frac{1/5}{(s-3)}$

$$h(t) = \frac{4}{5} e^{-2t} u(t) + \frac{1}{5} e^{3t} u(t)$$



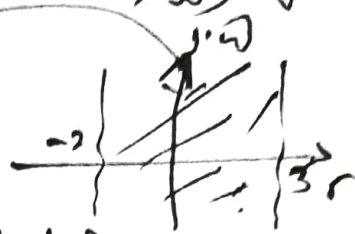
(3)

(i) If the system is stable

for the system ~~should~~ to be stable

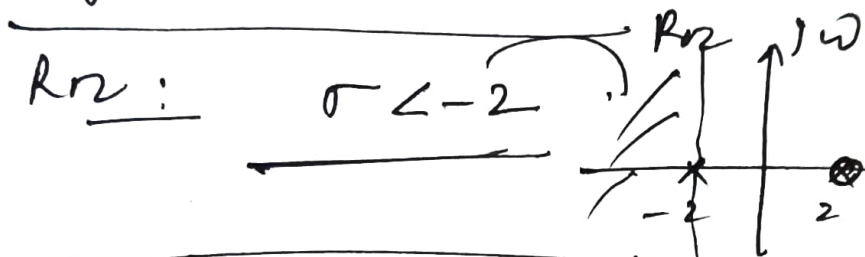
ROC should include the jw axis (imaginary axis)

So ROC: $-2 < \text{Re}(s) < 3$



$$h(t) = \frac{4}{5} e^{-2t} u(t) - \frac{1}{5} e^{3t} u(t-1)$$

(ii) If the system is Non Causal



$$\text{So } h(t) = -\frac{4}{5} e^{-2t} u(-t) - \frac{1}{5} e^{3t} u(-t)$$

Unilateral Laplace transform

$$\mathcal{L}^+[x(t)] = X^+(s) = \int_0^{\infty} x(t) e^{-st} dt$$

$$\rightarrow \mathcal{L}^+\left[\frac{dx}{dt}\right] = sX(s) - x(0^+)$$

$$\rightarrow \mathcal{L}^+\left[\frac{d^2x}{dt^2}\right] = s^2X(s) - sx(0^-) - \frac{dx(0^-)}{dt}$$

Solution of Differential Equations (output response of the system)
using Laplace transform

$$\text{Total Response} = \begin{array}{l} \text{(free response)} \\ \text{Natural response} \end{array} + \text{forced response}$$

$\downarrow$

due to initial condⁿ only
(all inputs are neglected)

$\downarrow$
due to inputs only
(all initial condⁿ are neglected)

Ques A system is described by the differential equation

$$\frac{d^2y(t)}{dt^2} + 5\frac{dy(t)}{dt} + 6y(t) = x(t)$$

- (a) find the impulse response of the system
- (b) for the initial conditions $\frac{dy(0^-)}{dt} = 2$, $y(0^-) = 1$ and input $x(t) = u(t)$, find the free and forced response of the system.
- (c) The total response of the system

Solⁿ:-

(5)

$$\frac{d^2 y(t)}{dt^2} + 5 \frac{dy(t)}{dt} + 6y(t) = x(t)$$

(a)

Taking Laplace transform on both side

$$s^2 y(s) + 5s y(s) + 6y(s) = X(s)$$

$$H(s) = \frac{Y(s)}{X(s)} = \frac{1}{s^2 + 5s + 6} = \frac{1}{(s+3)(s+2)}$$

Impulse response

$$h(t) = \mathcal{L}^{-1}(H(s)) = \mathcal{L}^{-1}\left[\frac{1}{(s+3)(s+2)}\right]$$

$$= \mathcal{L}^{-1}\left[\frac{1}{(s+2)} - \frac{1}{(s+3)}\right]$$

$$h(t) = e^{-2t} u(t) - e^{-3t} u(t)$$

(b)

Free response

Taking unilateral Laplace on both side by taking initial condⁿ and neglecting inputs

$$\frac{d^2 y}{dt^2} + 5 \frac{dy(t)}{dt} + 6y(t) = 0$$

$$s^2 y(s) - s y(0^-) - \frac{dy(0^-)}{dt} + 5[s y(s) - y(0^-)] + 6y(s) = 0$$

$$\Rightarrow (s^2 + 5s + 6) y(s) = s y(0^-) + 5 y(0^-) + \frac{dy(0^-)}{dt}$$
$$= (s+5) + 2 = (s+7)$$

$$y(s) = \frac{(s+7)}{s^2 + 5s + 6} = \frac{5}{(s+2)} - \frac{4}{(s+3)}$$

$$y_h(t) = 5 e^{-2t} u(t) - 4 e^{-3t} u(t)$$

⑥
→ Forced response

Neglecting the initial condⁿ and consider the input.

$$x(t) = u(t) \Rightarrow X(s) = \frac{1}{s}$$

$$\frac{d^2 y(t)}{dt^2} + 5 \frac{dy(t)}{dt} + 6y(t) = x(t)$$

$$\Rightarrow s^2 Y(s) + 5s Y(s) + 6 = X(s) = \frac{1}{s}$$

$$Y(s) = \frac{1}{s(s^2 + 5s + 6)} = \frac{1}{s(s+3)(s+2)}$$

$$= \frac{1}{6} \cdot \frac{1}{s} - \frac{1}{2(s+2)} + \frac{1}{3(s+3)}$$

$$\boxed{y_f(t) = \frac{1}{6} u(t) - \frac{1}{2} e^{-2t} u(t) + \frac{1}{3} e^{-3t} u(t)}$$

© Total response of the system:

$$y(t) = y_n(t) + y_f(t)$$

$$= 5e^{-2t} u(t) - 4e^{-3t} u(t) + \frac{1}{6} u(t) - \frac{1}{2} e^{-2t} u(t) + \frac{1}{3} e^{-3t} u(t)$$

$$\boxed{y(t) = \frac{1}{6} u(t) + \frac{9}{2} e^{-2t} u(t) - \frac{11}{3} e^{-3t} u(t)}$$

(7)

Ques. Using Laplace transform, solve the following differential Equation.

$$\frac{d^2 y(t)}{dt^2} + y(t) = x(t)$$

Given $\frac{dy(0^-)}{dt} = 2$, $y(0^-) = 1$, for input $x(t) = \cos t$

Solⁿ:-

$$\frac{d^2 y(t)}{dt^2} + y(t) = x(t)$$

Taking ^{unilateral} Laplace on both side

$$s^2 y(s) - s y(0^-) - \frac{dy(0^-)}{dt} + y(s) = X(s)$$

$$s^2 y(s) - s(1) - 2 + y(s) = \frac{s}{s^2 + 4}$$

$$y(s) [s^2 + 1] = \frac{s}{s^2 + 4} + s + 2$$

$$y(s) = \frac{s}{(s^2 + 4)(s^2 + 1)} + \frac{(s + 2)}{(s^2 + 1)}$$

$$= \frac{\cancel{5} s + (s + 2)(s^2 + 4)}{(s^2 + 1)(s^2 + 4)}$$

$$= \frac{s^3 + 2s^2 + 5s + 8}{(s^2 + 1)(s^2 + 4)} = \frac{As + B}{(s^2 + 1)} + \frac{Cs + D}{(s^2 + 4)}$$

$$= \frac{\frac{4}{3}s + 2}{(s^2 + 1)} + \frac{(-\frac{1}{3}s)}{s^2 + 4}$$

$$Y(s) = \frac{\frac{1}{3}s}{s^2+1} + \frac{2(1)}{s^2+1} - \frac{1}{3} \frac{s}{s^2+4}$$

⑧

Take Inverse Laplace

$$y(t) = \frac{1}{3} \cos t \cdot u(t) + 2 \sin t \cdot u(t) - \frac{1}{3} \cos 2t \cdot u(t)$$